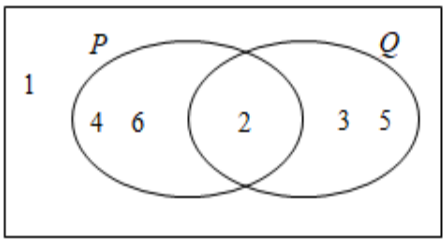
Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	53 035	**1**	
1(b)	8400	**1**	
2(a)	Acute	**1**	
2(b)(i)	Chord	**1**	
2(b)(ii)	16	**1**	
3	$\frac{1}{10}$	**1**	
4(a)	14	**1**	
4(b)	3375	**1**	
5(a)	$16 \div (8 + 4 \times 2) = 1$	**1**	
5(b)	$(16 \div 8 + 4) \times 2 = 12$	**1**	
6	$\frac{114}{360} \times 840$	**1**	
7	5300	**1**	
8(a)	108	**2**	**B1** for 9 cm
8(b)	Accurate position of town *C* with arcs	**3**	**B2** for accurate position of town *C* without arcs or two arcs, one accurate or **B1** for 6 and 8 soi
9	2	**1**	
10	285	**2**	**M1** for 180 + 105 or 75 or 105 seen in correct position at *B*
11(a)	Any square number greater than 10	**1**	
11(b)	Any irrational number	**1**	
12(a)	(3, 1)	**1**	

Question	Answer	Marks	Partial Marks
12(b)	D plotted at (–2, –1)	**1**	
12(c)	E plotted at (1, –2)	**2**	**B1** for E plotted at (1, k) or (k, –2) or $\overrightarrow{AE} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$
13	76.25 76.35	**2**	**B1** for each or both correct and reversed
14	$x + x + 8 + 2x - 3 = 117$ or better	**M2**	or **B1** for $x + 8$ or $2x - 3$
	$4x + 5 = 117$ oe or better	**M1**	
	28	**A1**	If 0 scored, **SC1** for the correct answer with no algebra
15	45	**2**	**M1** for $\frac{11+7}{2} \times 5$ oe
16(a)	Intersection shaded	**1**	
16(b)		**2**	**B1** for four or five numbers in the correct place
17	0.0625	**1**	
18	$\frac{7}{4}$ $\frac{9}{12}$	**B1**	
	$\frac{21}{12}$ $1 - \frac{2}{12}$	**M1**	
	$\frac{5}{6}$ $\frac{5}{6}$	**A1**	
19	8	**2**	**M1** for $\frac{5-4.60}{5}$ [×100] or $\frac{4.60}{5} \times 100$
20	$2t^4$	**2**	**B1** for $2t^n$ or kt^4 ($n,k \neq 0$)
21(a)	4.5×10^4	**1**	
21(b)	0.0206	**1**	
22(a)	1 2 4 7 14 28	**2**	**B1** for 4 or 5 correct with no extras or 6 correct with one extra
22(b)	$2 \times 3 \times 3 \times 3$ or 2×3^3	**2**	**M1** for 2, 3, 3, 3 or correct factor tree